

Pricing of a novel parametric weather derivative - A Multi-variate Spatial Basket

HKUST

1. A Local Risk Exposure - Baseline Stochastic Models Fitted to local Data: $S = f(Y_l(t))$
 - ▶ Weather variables and Climate events - Class of Gauss-Markov processes suitable for indifference pricing measure.
 - ▶ temp - Continuous-time AR(p) model, Seasonal volatility fit.
 - ▶ wind spd anomalies - Static GEV or Normal Inverse Gaussian fit (Generalized Hyperbolic Distribution)
 - ▶ wind dir and spd - Dynamic Multivariate Ornstein Uhlenbleck fit
2. A Global Traded Index - Futures / Parametric Proxy: $Y_g(t)$
 - ▶ Constructing Real Estate investment portfolio and HDD, CDD, CAT futures index proxy under risk-neutral distribution.
 - ▶ Climate event data for Compound Risk Indices.
3. Pricing + Hedging - Spatial Basket: $X(t)$ and $a(t)$ Strategy
 - ▶ Constructing synthetic future index.
 - ▶ Spatio-temporal climate model - params from literature

Biovariables and Morphology params

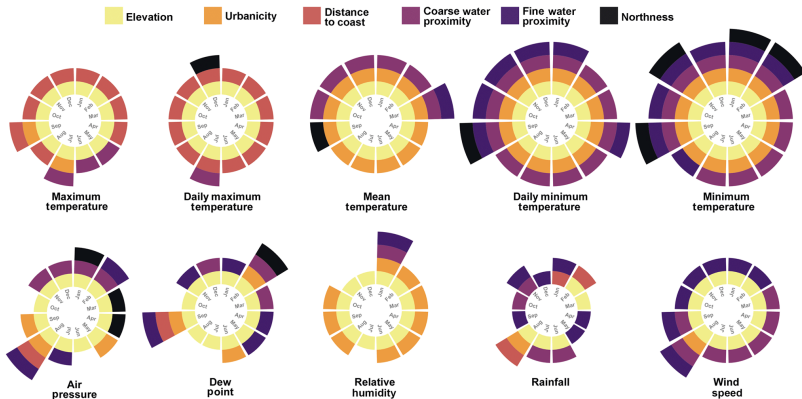
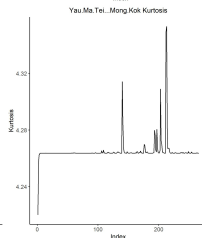
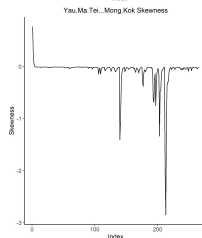
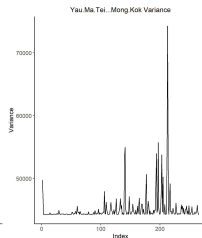
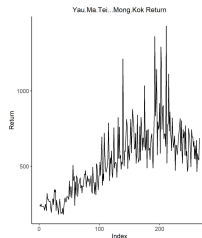
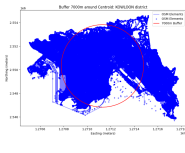
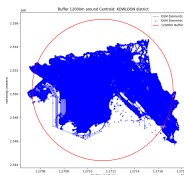
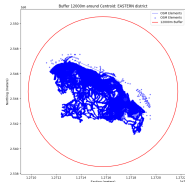
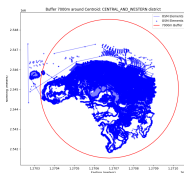


Figure: Reference variables

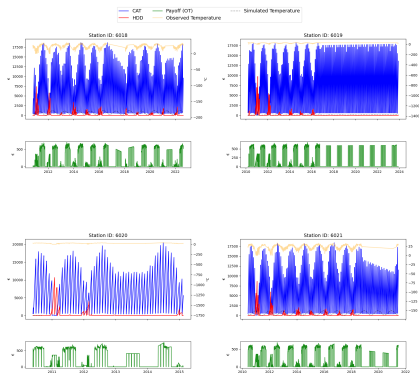
Real Estate Investment and Weather Index - Y_l ; Y_g

- **Exposure Model** (since no market portfolio)
- Real Estate - Estate-wise aggregation: local climate exposure Y_l



Real Estate Investment and weather index - Y_I ; Y_g

Futures on HDD, CDD, CAT, and PRIM - usual traded indices.
Multivariate Index strategy based on a linearization based feedback law : get synthetic global climate index Y_g



Real Estate Investment and Weather Index - Y_I and Y_g

► RE Financial Exposure:

$$\text{Investment Portfolio} = \sum_{i=1}^k P_i$$

where P_i represents the estate-wise time series of price.

► Local Climate Risk Exposure: climate index for the i^{th} estate:

$$Y_I = \sum_{i=1}^k (a + b \cdot \log(\text{Local Climate Index}_i) + \epsilon_i)$$

► Global Climate Risk Hedge: weight assigned to the j^{th} district

$$Y_g = \sum_{j=1}^m w_j \cdot \text{District Climate Index}_j$$

► Cauchy Problem: Dynamics between Y_g and Y_I

Fitted Parameters - Wind GEV

Table: Extreme Value model params - peaks over threshold analyses of Waglan Island typhoon data (1953-2006)

Parameter	200 m		50 m	
	20 m/s threshold	24 m/s threshold	20 m/s threshold	24 m/s threshold
Rate (/year), λ	1.73	1.69	1.73	1.69
Scale, σ (m/s)	6.8	9.1	6.2	8.5
Shape, k	0.041	0.031	0.0*	0.0*
20 year R.P. (m/s)	42.3	54.2	42.0	53.9
50 year R.P. (m/s)	47.5	61.5	47.7	61.6
200 year R.P. (m/s)	55.1	72.2	56.3	73.4
1000 year R.P. (m/s)	63.4	84.1	66.3	87.0
$(V_{1000}/V_{50})^2$	1.78	1.87	1.93	1.99
$(V_{1000}/V_{50})^{2.5}$	2.06	2.19	2.28	2.37

Source - Holmes et al

Fitted Parameters - Multivariate OU

Table 1: Multivariate Ornstein-Uhlenbeck Model Parameters

Long-Term Mean Vector (μ)		
Variable		
speed	3.4932	
direction	145.4068	
Drift Matrix (Θ)		
Variable	Θ_1	Θ_2
speed	0.0617	0.0008
direction	0.9188	0.1953
Diffusion Matrix (Σ)		
Variable	Σ_{11}	Σ_{12}
speed	0.9110	2.7846
direction	2.7846	1813.5325

Figure: station 6001 (The Hong Kong Polytechnic University (PolyU)) - 10 years data based

Spatial Hedging - Weight Matrix - Global index Y_g

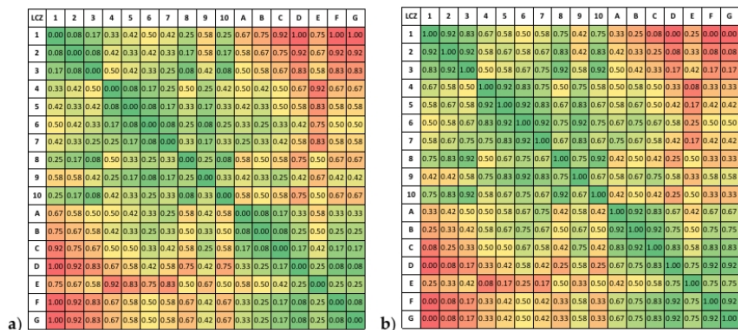


Figure 3. (a) Dissimilarity and (b) similarity metric for LCZ classes (Appendix A in [4]).

Figure: LCZ (Local Climate Zone) based weights matrix reference

Spatial Hedging - Weight Matrix - Global index Y_g

- ▶ Model with no spatial dependency in Λ , α , and σ .
- ▶ spatial model for index construction - IDW, ordinary kregging, regression kregging - to add: EDA corrlle graphs
- ▶ Spatial dependency modeled by a spherical correlation function (Benth 2012)

$$q(x, y) = 1 - \frac{3}{2} \frac{|x - y|}{\gamma} + \frac{1}{2} \frac{|x - y|^3}{\gamma^3}$$

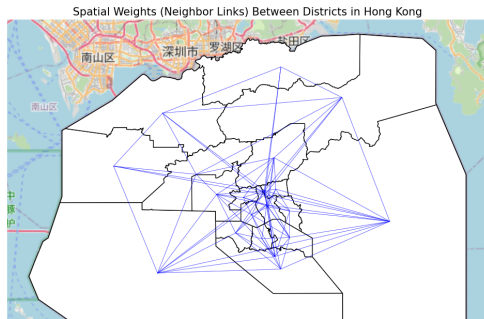


Figure: Rent Roll Based Network

Pricing and Model Specification - no basis risk

$$F(t, y) \left(\alpha g(t, y) + \frac{1}{2} \frac{\mu^2(y)}{\sigma^2(y)} \right) = \frac{\partial F}{\partial t}(t, y) + GF$$

with terminal condition $F(T, y) = 1$. We decompose the state variable Y by using proxies of y : Y_1 as local risk and Y_2 as global risk (for payoff). Using Feynman-Kac -

$$F(t, Y_1, Y_2) = \mathbb{E}_y \left[\exp \left(- \int_t^T \left(\alpha g(s, Y_2) + \frac{1}{2} \frac{\mu^2(Y_1(s))}{\sigma^2(Y_1(s))} \right) ds \right) \right]$$

E_y - over joint distribution Y_1 and Y_2 s.t. inf generator G accounts for coupling terms with $V(t, x, y) = -e^{-\alpha x} F(t, Y_1, Y_2)$

General Solution (Carmona 2005) -

$$p = \frac{1}{\alpha} \ln \left(\frac{\mathbb{E}_y \left[\exp \left(- \int_0^T \frac{1}{2} \frac{\mu^2(y_s)}{\sigma^2(y_s)} ds \right) \right]}{\mathbb{E}_y \left[\exp \left(- \int_0^T \left(\alpha g(s, \hat{Y}_s) + \frac{1}{2} \frac{\mu^2(y_s)}{\sigma^2(y_s)} \right) ds \right) \right]} \right)$$

Pricing and Model Specification - basis risk

Unhedgable basis/residual risk or when coupling effect is negligible/can be treated as a perturbation to the independent case -

With terminal condition $F(T, Y_1, Y_2) = 1$

$$F(t, Y_1, Y_2) \left(\alpha g(t, Y_2) + \frac{1}{2} \frac{\mu^2(Y_1)}{\sigma^2(Y_1)} \right) = \frac{\partial F}{\partial t}(t, Y_1, Y_2) + \mathcal{L}_{Y_1} F + \mathcal{L}_{Y_2} F$$

$$\begin{aligned} F(t, Y_1, Y_2) &= \mathbb{E}_{Y_1, Y_2} \left[\exp \left(- \int_t^T \left(\alpha g(s, Y_2(s)) + \frac{1}{2} \frac{\mu^2(Y_1(s))}{\sigma^2(Y_1(s))} \right) ds \right) \right] \\ &= \mathbb{E}_{Y_1} \left[\exp \left(- \int_t^T \frac{1}{2} \frac{\mu^2(Y_1(s))}{\sigma^2(Y_1(s))} ds \right) \right] \times \mathbb{E}_{Y_2} \left[\exp \left(- \int_t^T \alpha g(s, Y_2(s)) ds \right) \right] \end{aligned}$$

Trading strategy: $\phi_t = \frac{\mu(Y_1)}{(\sigma^2(Y_1)\alpha)}$ — Case ignored since $p = 0$

Pricing and Model Specification - basis risk

Simplified coupled case (b/w local and global risk) - Price through the 'distortion solution' -

$$p = \frac{\ln \mathbb{E} \left[e^{(1-\rho^2)\Psi(Y_2(T))} \frac{dQ}{dP} \right]}{\alpha(1-\rho^2)}, \quad \frac{dQ}{dP} = e^{-\frac{1}{2} \int_0^T (\mu_1^2/\sigma_1^2) d\tau + \int_0^T \frac{\mu_1}{\sigma_1} dW_t^S}$$

$$\phi_t = \frac{\rho}{\sigma(Y_1)} a(Y_2) p' + \frac{\mu(Y_1)}{\sigma(Y_1)^2 \alpha}.$$

where -

$$dS_t = \mu(Y_1, t) dt + \sigma(Y_1, t) dW_t^S$$

$$dY_t = b(Y_2, t) dt + a(Y_2, t) (\rho dW_t^S + \sqrt{1-\rho^2} dW_t^{S\perp})$$

$$W^{S\perp} = \frac{1}{\sqrt{1-\rho^2}} W^Y - \frac{\rho}{\sqrt{1-\rho^2}} W^S$$

On-hold

coupling dynamics - adv-diff-reac system

$$\frac{\partial y_1}{\partial t} + w \cdot \nabla y_1 - \nabla \cdot (\epsilon \nabla y_1) = f_1 - K y_1 y_2$$

$$\frac{\partial y_2}{\partial t} + w \cdot \nabla y_2 - \nabla \cdot (\epsilon \nabla y_2) = f_2 - K y_1 y_2$$

$$\frac{\partial u}{\partial t} + w \cdot \nabla u - \nabla \cdot (\epsilon \nabla u) = f_3 - K y_1 y_2 - K u$$

$A + B \rightarrow C$.

Temporal Dynamics of State Variables

Temperature example - for basis risk as the source of market incompleteness

$$dT(t, x) = d\Lambda(t, x) - \alpha(x)(T(t, x) - \Lambda(t, x)) dt + \sigma(t, x) dW(t, x)$$

- ▶ $W(t, \cdot)$: $L^2(D)$ -valued Wiener process
- ▶ Continuous spatial covariance function $q(x, y)$, strictly positive definite and symmetric.
- ▶ Given weather index $I(x_i)$ at locations $x_1, \dots, x_n$.
- ▶ Optimal strategy $a(t)$ minimizing variance of error in replicating $I(y)$ at location y using spatial correlation and geometry of locations.
- ▶ Risk Neutral Time-Dynamics of Index Futures Prices;

$$dF_{\text{index}}(t, \tau_1, \tau_2) = \sigma(t) \int_{\tau_1}^{\tau_2} e_1' \exp(A(s - t)) e_p \Phi$$

Φ is the cumulative standard normal distribution

Fitting the Model

- Explicit solution of $X(t)$:

$$X(s) = \exp(A(s - t))x + \int_t^s \exp(A(s - u))e_p\sigma(u) dB(u)$$

- Temperature $T(t)$ is defined as:

$$T(t) = \Lambda(t) + X_1(t)$$

- $X_1(t)$: CAR(p) model, $\Lambda(t)$: seasonality function

1. Fit seasonal function $\Lambda(t)$ with least squares.
2. Fit AR(p) model on deseasonalized temperatures.
3. Fit seasonal volatility $\sigma(t)$ to residuals.

Fitting the Model

Continuous time OU fitted density

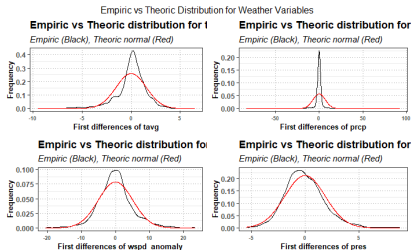


Figure: weather variables at station 6001 - fitted density

Fitting the Model

Generalised extreme value distribution : CDF for maximum wind speed

$$F_U(U) = \begin{cases} \exp \left\{ - \left[1 + k \frac{U-u}{a} \right]^{-1/k} \right\}, & k \neq 0, 1 + k \frac{U-u}{a} > 0, \\ \exp \left\{ - \exp \left[- \frac{U-u}{a} \right] \right\}, & k = 0. \end{cases} \quad (1)$$

k : shape parameter (tail behavior), a : scale parameter (controls the spread), u : location parameter (shifts).

- ▶ Type II (Fréchet) Distribution: When $k < 0$, Type III (Weibull) Distribution: When $k > 0$
- ▶ Type I (Gumbel) Distribution: As $k \rightarrow 0$ - directly fit Gumbel distribution to the annual maxima

Contract Payoff and Pricing

Utility with a derivative written on the underlying process Y - akin to Asian structure and $K = 0$

- ▶ Cumulative exposure - $f_1(Y) = \left(\int_0^T Y_t dt - K \right)^+$
- ▶ Duration - $f_2(Y) = \left(\int_0^T \mathbb{1}_{\{Y_t > \epsilon\}} dt - K \right)^+$,

$$f(Y) = \int_0^T h(Y_s) ds, \quad 0 \leq t' \leq t \leq T$$

- ▶ To Show - Numerical solution of underlying optimization problem analysed through its dual under HJB (David)
- ▶ To Show - Explicit closed form solution to HJB: RN measure vs historical measure vs Indifference measure
- ▶ To Show - RE investment variance with and without the composite option